CLASS 10 — MATHS · HARD LEVEL CHAPTER TEST SET 1

Chapter 5: **Arithmetic Progressions** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Hard level: application and higher-order questions.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. If the 4th term of an AP is 9 and the 9th term is 4, then the 13th term is: 1

(a) 36 **(b)** 0 **(c)** 13 **(d)** 1

2. The sum of the first 13 terms of an AP is 390, and its first term is 6. The 13th term is: 1

(a) 24 **(b)** 60 **(c)** 66 **(d)** 54

3. In an AP, $S_8 = 200$ and $S_7 = 154$. Then $a_8 =$ 1

(a) 40 **(b)** 46 **(c)** 52 **(d)** 354

4. The sum of the first n terms of an AP is $S_n = 2n^2 + 3n$. Its common difference is: 1

(a) 2 **(b)** 3 **(c)** 4 **(d)** 5

5. The sum of the series $5 + 9 + 13 + \cdots + 101$ is: 1

(a) 2650 **(b)** 1224 **(c)** 1430 **(d)** 1325

6. How many terms of the AP $20, 16, 12, 8, \ldots$ give sum 0? 1

(a) 11 **(b)** 10 **(c)** 22 **(d)** 12

7. If the 5th term of an AP is 11 and the 11th term is 5, then the 16th term is: 1

(a) 55 **(b)** 1 **(c)** 16 **(d)** 0

8. The sum of the first 9 terms of an AP is 279, and its first term is 3. The 9th term is: 1

(a) 28 **(b)** 59 **(c)** 65 **(d)** 62

9. **Assertion (A):** The list $3, 7, 11, 15, \ldots$ forms an AP. 1

Reason (R): In an AP the difference between any two consecutive terms is the same constant.

Choose the correct option:

(a) Both A and R are true and R is the correct explanation of A.

(b) Both A and R are true but R is not the correct explanation of A.

(c) A is true but R is false.

(d) A is false but R is true.

10. **Assertion (A):** The 10th term of the AP $2, 5, 8, \ldots$ is 29. 1
Reason (R): The nth term of an AP with first term a and common difference d is $a_n = a + nd$.
Choose the correct option:
(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

SECTION — B (Very Short Answer Type) — 4 × 2 = 8

Answer in about 20–30 words each.

11. The 4th term of an AP is 9 and the 9th term is 4. Find the 13th term. 2

12. If $S_n = 3n^2 + 4n$, find the first term and the common difference. 2

13. In an AP, $a_3 + a_7 = 34$ and $a_2 + a_9 = 37$. Find a and d. 2

14. The 5th term of an AP is 12 and the 12th term is 5. Find the 17th term. 2

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 40–60 words each.

15. In an AP, $S_6 = 75$ and $S_{10} = 185$. Find the AP. 3

16. Find the sum of the first 16 terms of the AP whose 3rd term is 14 and 8th term is 34. 3

17. Find the sum of all integers between 10 and 200 that leave remainder 3 when divided by 7. 3

OR

In an AP, $S_5 = 50$ and $S_9 = 162$. Find the AP. 3

18. Find the sum of the first 20 terms of the AP whose 2nd term is 14 and 7th term is 39. 3

OR

Find the sum of all integers between 20 and 300 that leave remainder 4 when divided by 9. 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

19. **Case study.** Mohan starts a monthly savings plan. He saves ₹ 100 in the first month and increases his saving by ₹ 20 every month. 5
(i) Write the first term and the common difference of the AP formed by his monthly savings.
(ii) How much does he save in the 10th month?
(iii) Find his total savings in the first 12 months.
(iv) In which month will he save ₹ 400, and what will his total savings be up to that month?

20. A ladder has 31 rungs whose lengths decrease uniformly from 90 cm at the bottom to 30 cm at the top. Find the length of the wood needed for the rungs and the length of the middle rung. 5

OR

A loan is repaid in 40 monthly instalments; the first is ₹600 and each later instalment is ₹45 more than the previous. Find the instalment in the last month and the total amount repaid. 5

END OF QUESTION PAPER — **Total: 40 Marks**